

# V6.2 · Information Bottleneck Optimisation

What abstraction costs, measured

|           |                                                                                                                                                                                                                                                         |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective | Replace top-1 accuracy as the optimisation objective with a direct measurement of the information retained by each level of abstraction.                                                                                                                |
| Method    | For $K = 2 \dots 17$ the motif activations are projected to $K$ themes and reconstructed by least squares. Explained variance, reconstruction error, KL divergence, mutual information with the chemical family and the compression ratio are recorded. |
| Frozen    | Atlas 09ed804a40836f4a05a91ba10900cded; the V6 MSS map $M$ is an input, never rewritten.                                                                                                                                                                |
| Reproduce | results/v6_rebuild/code/p10 → p11 → p12 → p13                                                                                                                                                                                                           |

Parts 5 & 9 — the information bottleneck: abstraction has a measurable price

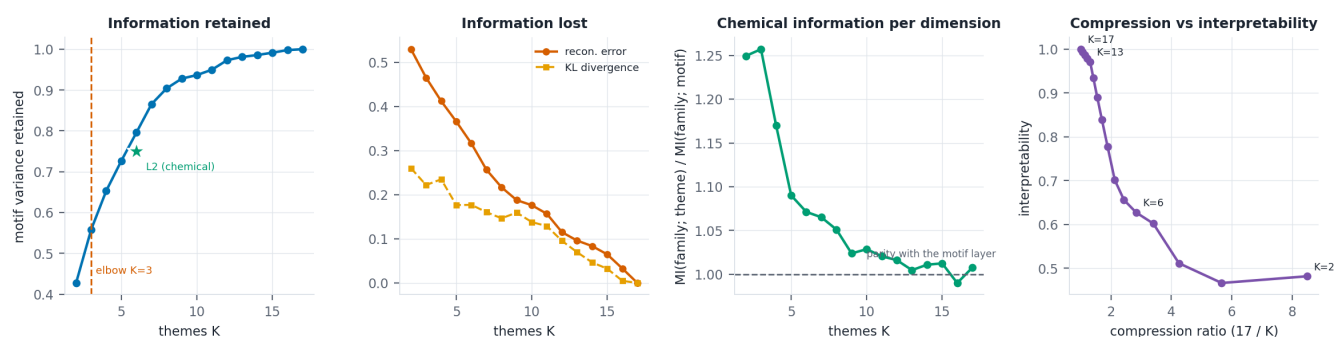


Figure 1 — Information retained, information lost, chemical information per dimension, and compression against interpretability.

## Results

| K  | variance retained | recon. error | KL     | MI retained | compression | adm. |
|----|-------------------|--------------|--------|-------------|-------------|------|
| 2  | 0.4275            | 0.5295       | 0.2601 | 1.2493      | 8.50        |      |
| 3  | 0.5582            | 0.4651       | 0.2218 | 1.2570      | 5.67        |      |
| 4  | 0.6533            | 0.4120       | 0.2349 | 1.1702      | 4.25        |      |
| 5  | 0.7268            | 0.3657       | 0.1765 | 1.0901      | 3.40        |      |
| 6  | 0.7961            | 0.3160       | 0.1770 | 1.0713      | 2.83        |      |
| 7  | 0.8656            | 0.2565       | 0.1604 | 1.0651      | 2.43        |      |
| 8  | 0.9041            | 0.2167       | 0.1465 | 1.0511      | 2.12        |      |
| 9  | 0.9280            | 0.1877       | 0.1595 | 1.0239      | 1.89        |      |
| 10 | 0.9366            | 0.1762       | 0.1380 | 1.0287      | 1.70        |      |
| 11 | 0.9498            | 0.1567       | 0.1288 | 1.0206      | 1.54        |      |
| 12 | 0.9730            | 0.1151       | 0.0955 | 1.0162      | 1.42        |      |
| 13 | 0.9812            | 0.0961       | 0.0698 | 1.0048      | 1.31        | yes  |
| 14 | 0.9857            | 0.0836       | 0.0466 | 1.0110      | 1.21        | yes  |
| 15 | 0.9914            | 0.0648       | 0.0324 | 1.0125      | 1.13        | yes  |

| K  | variance retained | recon. error | KL     | MI retained | compression | adm. |
|----|-------------------|--------------|--------|-------------|-------------|------|
| 16 | 0.9978            | 0.0329       | 0.0050 | 0.9901      | 1.06        | yes  |
| 17 | 1.0000            | 0.0000       | 0.0000 | 1.0077      | 1.00        | yes  |

**The V6.2 L2 layer (K=6) retains 0.751 of the motif variance at a 2.83× compression** — against 0.981 at V6's K=13 with only 1.31× compression. V6.2 gives up 0.23 of motif variance to more than double the abstraction.

### Two results worth stating precisely

- **The elbow sits at K=3** by maximum curvature of the variance curve — but at that point only 0.56 of the motif variance survives. The elbow is a mathematical feature of the curve, not a recommendation.
- **Mutual information per dimension is higher in the theme space than in the motif space for  $K \leq 13$**  (MI retained > 1.0). This is not free information: MI is normalised per dimension, so fewer dimensions each carry more chemical-family information. Abstraction concentrates the signal even as it discards variance.

## Interpretation

The two curves answer different questions. Reconstruction variance says how much of the motif pattern could be rebuilt — it falls steeply below  $K \approx 8$ . Mutual information says how much chemistry the representation still distinguishes — it stays near or above parity all the way down to  $K=3$ . A hierarchy chosen on variance alone would keep 13+ themes; one chosen on chemical information could go to 4. V6.2 sits between them at 6 because that is where the grouping is also chemically nameable.

## Limitations

- MI is estimated by per-dimension quantile binning; it is a plug-in estimate on 167 analytes and should be read as a trend, not a precise nat count.
- Reconstruction uses an unconstrained least-squares inverse, which is generous — a non-negative inverse would retain less.
- The sweep uses the hybrid clustering; the chosen L2 grouping is not on that curve and is reported as a separate labelled point.

## Recommendations

- Report both variance retained and MI retained whenever a hierarchy level is proposed; they disagree, and the disagreement is the interesting part.
- Use  $K=6$  for interpretation and  $K=17$  (the motif layer) whenever information matters more than nameability.

Atlas 09ed804a40836f4a05a91ba10900cded verified unchanged. All values recomputed from results/v6\_rebuild/.